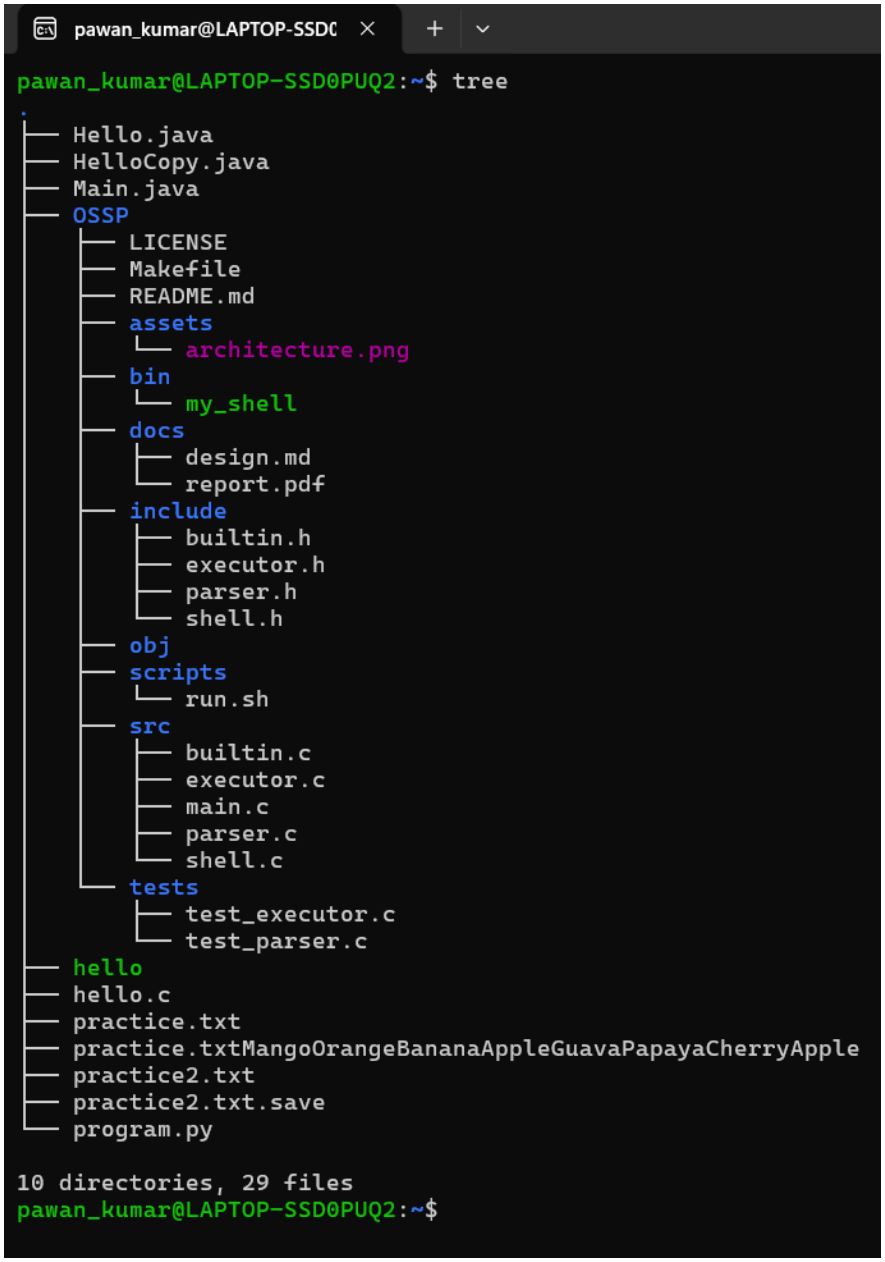
WEEK1 SKILL

Session1:

1. **To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile.**

Task -1: Create Project Structure as follows:
Note: For creating this structure use mkdir and touch commands.



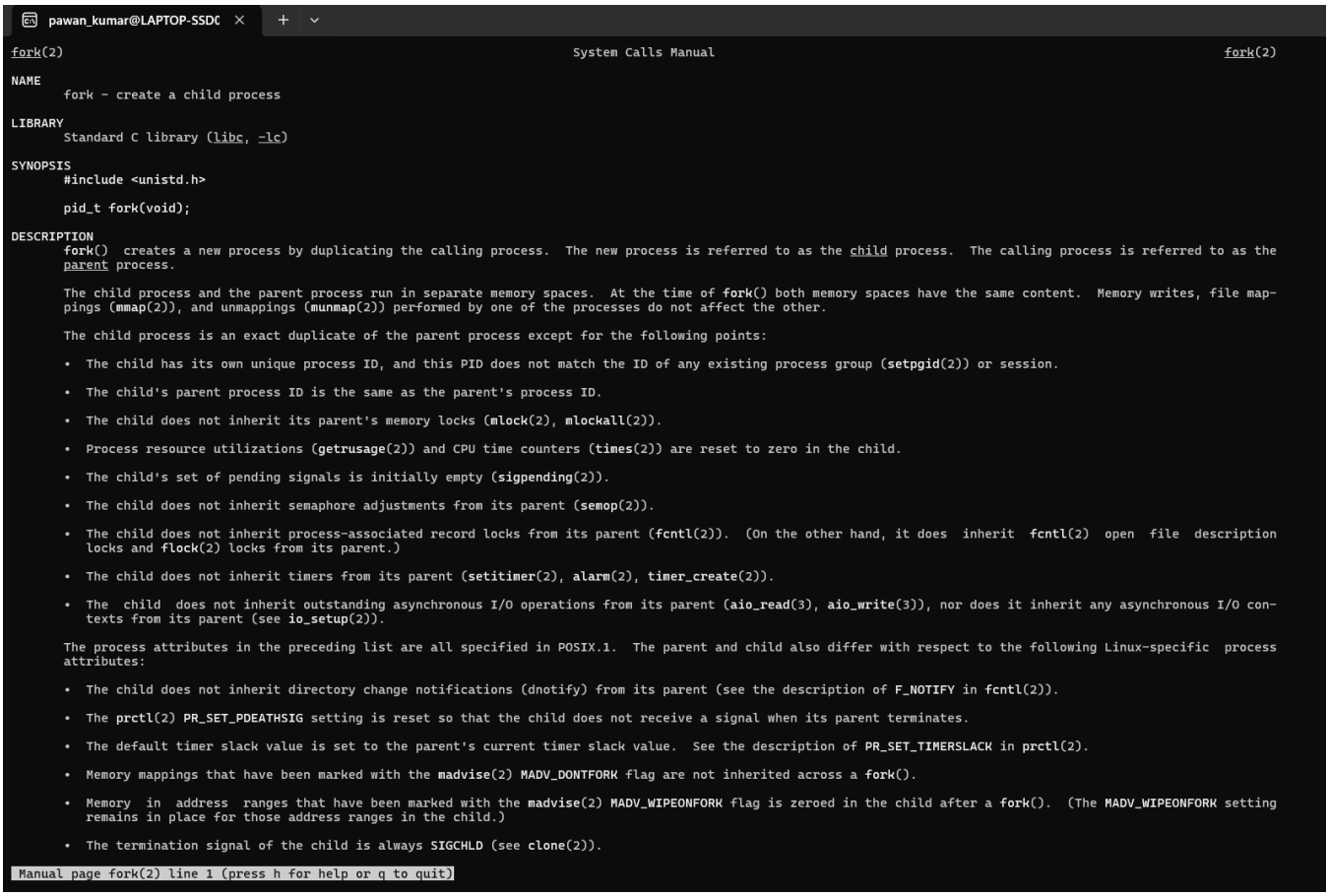
```
 GNU nano 8.7.1                                task2.c *
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();   // Create a new process

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child Process
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        // Parent Process
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~$ gcc task2.c -o task2
pawan_kumar@LAPTOP-SSD0PUQ2:~$ ./task2
This is the Parent Process.
Parent PID = 1019
Child PID = 1020
This is the Child Process.
Child PID = 1020
pawan_kumar@LAPTOP-SSD0PUQ2:~$
```

Report: I created the necessary folders and files, initialized a Git repository, and verified the project structure using the tree command. Thus, I successfully completed the task and understood how to organize a project in Linux.

To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

Task-1: Using a manual command understand the fork() and exec() system calls



Fork()

Man exec:

```
exec(3)                  Library Functions Manual                  exec(3)

NAME
       execl, execlp, execle, execv, execvp, execvpe - execute a file

LIBRARY
       Standard C library (libc, -lc)

SYNOPSIS
       #include <unistd.h>

       extern char **environ;

       int execl(const char *path, const char *arg, ...
                       /*, (char *) NULL */);
       int execlp(const char *file, const char *arg, ...
                       /*, (char *) NULL */);
       int execle(const char *path, const char *arg, ...
                       /*, (char *) NULL, char *const envp[] */);
       int execv(const char *path, char *const argv[]);
       int execvp(const char *file, char *const argv[]);
       int execvpe(const char *file, char *const argv[], char *const envp[]);

   Feature Test Macro Requirements for glibc (see feature_test_macros(7)):

       execvpe():
           _GNU_SOURCE

DESCRIPTION
       The  exec()  family  of functions replaces the current process image
       with a new process image.  The functions described  in  this  manual
       page  are layered on top of execve(2).  (See the manual page for ex-
       ecve(2) for further details about the  replacement  of  the  current
       process image.)

       The  initial argument for these functions is the name of a file that
       is to be executed.

       The functions can be grouped based  on  the  letters  following  the
       "exec" prefix.
```

Report: I used the manual commands man fork and man exec to understand the fork() and exec() system calls. I read the manual pages and learned that fork() creates a new child process, while exec() replaces the current process with a new program. Thus, I successfully understood the purpose and working of both system calls.

Task-2: Write a C program to create a new process and accepting the user commands using fork() and exec() system calls.

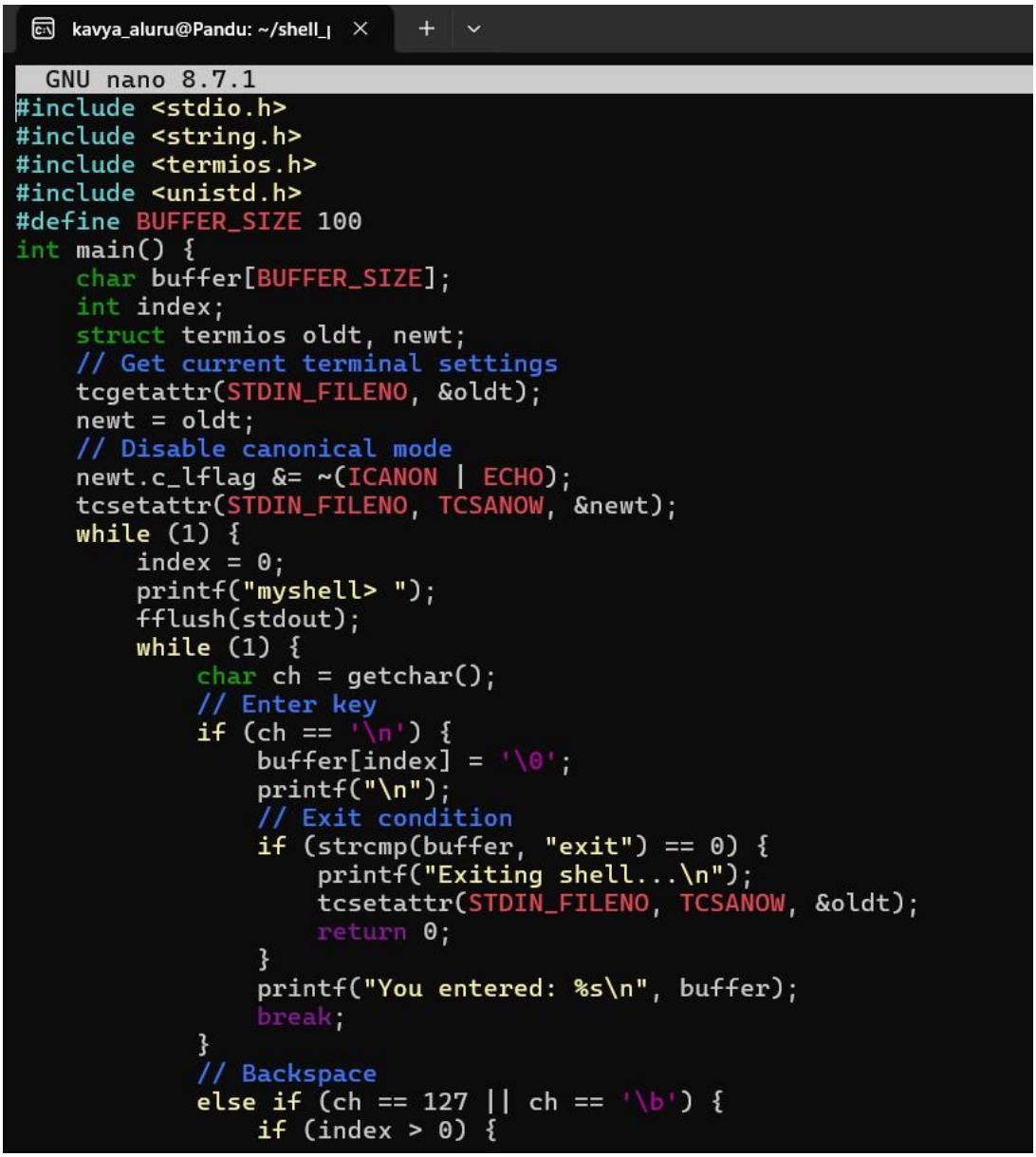
```
  GNU nano 8.7.1                                task2.c
#include <stdio.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();   // Create a new process

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child Process
        printf("This is the Child Process.\n");
        printf("Child PID = %d\n", getpid());
    }
    else {
        // Parent Process
        printf("This is the Parent Process.\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
```

```
pawan_kumar@LAPTOP-SSD0PUQ2:~/OSSP/task2$ gcc task2.c -o task2
pawan_kumar@LAPTOP-SSD0PUQ2:~/OSSP/task2$ ./task2
This is the Parent Process.
This is the Child Process.
Child PID = 2884
Parent PID = 2883
Child PID = 2884
pawan_kumar@LAPTOP-SSD0PUQ2:~/OSSP/task2$
```

Report: I wrote and executed a C program using the fork() system call to create a child process. I compiled and ran the program successfully and observed the creation of both the parent and child processes along with their Process IDs (PIDs). Thus, I understood the working of the fork() system call and the parent-child process relationship successfully.

SESSION 2:

1. To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop



```
Exiting shell...
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ nano shell.c
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ gcc shell.c -o shell
./shell
myshell> hello
You entered: hello
myshell> test123
You entered: test123
myshell> exit
Exiting shell...
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$
```

Report: I compiled and ran the shell program and tested it with the inputs hello, test123, and exit. I observed that the shell correctly displayed the entered commands and exited when the exit command was given. Thus, I successfully implemented and tested the main loop, user input handling, and exit condition of the shell program.

2. To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define BUFFER_SIZE 100

int main() {
    char command[BUFFER_SIZE];

    while (1) {
        printf("myshell> ");
        fflush(stdout);

        // Read user input
        if (fgets(command, BUFFER_SIZE, stdin) == NULL) {
            break;
        }

        // Remove newline
        command[strcspn(command, "\n")] = '\0';

        // Exit condition
        if (strcmp(command, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        // Ignore empty input
        if (strlen(command) == 0) {
            continue;
        }

        // Create child process
        pid_t pid = fork();
```

Report: I wrote and executed a C program using fgets() to read user input, handled the newline character, implemented the exit condition, and created a child process using fork(). Thus, I successfully implemented the main shell loop, user input handling, exit condition, and process creation.

make

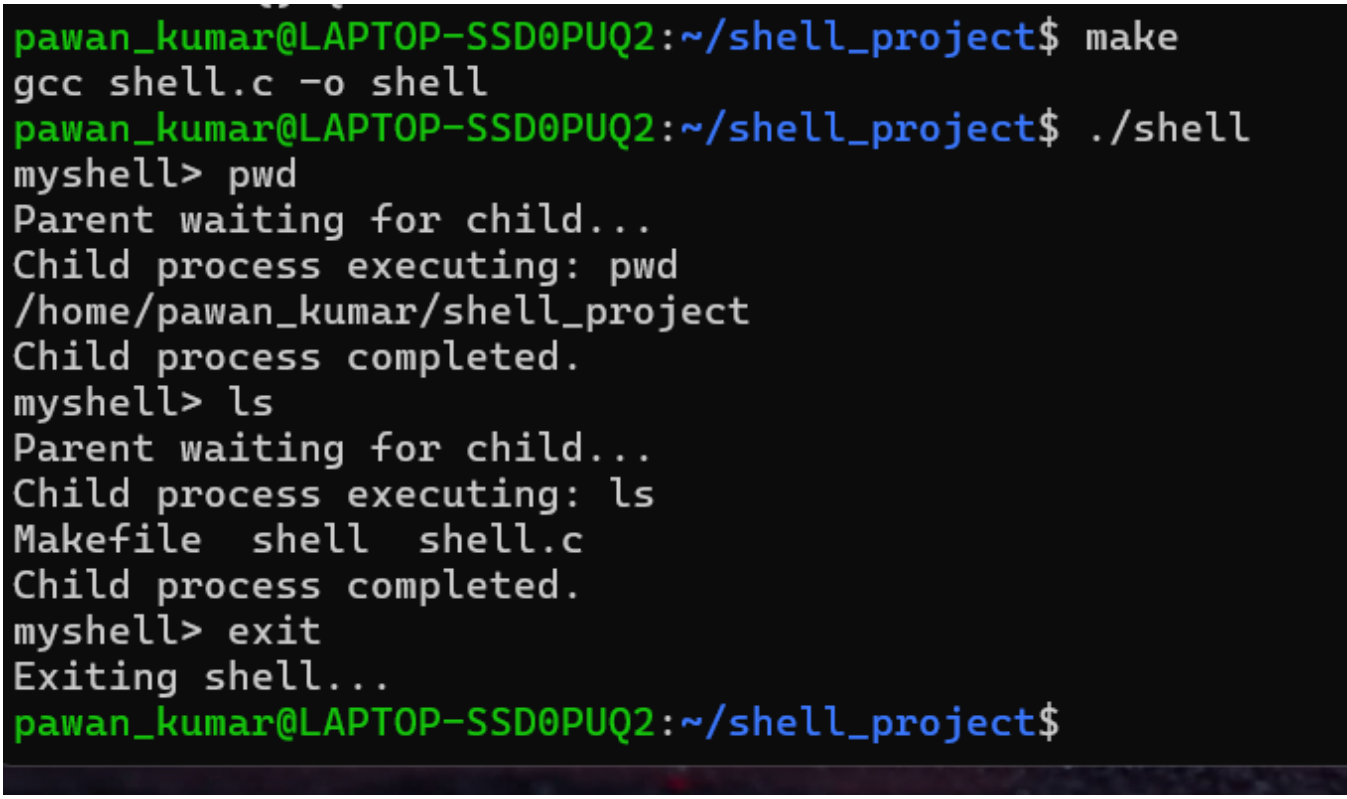
```
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ make
gcc shell.c -o shell
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ ./shell
myshell> exit
Exiting shell...
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ cd ~/shell_project
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$
```

Report:
I compiled the shell program using the make command and successfully executed it. I tested the exit command and observed that the shell displayed the exit message and terminated correctly. Thus, I successfully built and tested the shell program and verified its exit functionality.

```
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ make
make: 'shell' is up to date.
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$ ./shell
myshell> hello wor
You entered: hello wor
myshell> exit
Exiting shell...
pawan_kumar@LAPTOP-SSD0PUQ2:~/shell_project$
```

backspace

Report: I executed the shell program and tested it with a multi-character input, hello wor, and the exit command. I observed that the program correctly displayed the entered input and terminated successfully when exit was given. Thus, I successfully tested multi-character command input and the exit functionality of the shell.



fork()

Report: I compiled and executed the shell program and tested it with the pwd and ls commands. I observed that the shell successfully created child processes, executed the commands, and waited for the child processes to complete before accepting the next command. Thus, I successfully implemented and tested command execution using fork(), exec(), and wait().